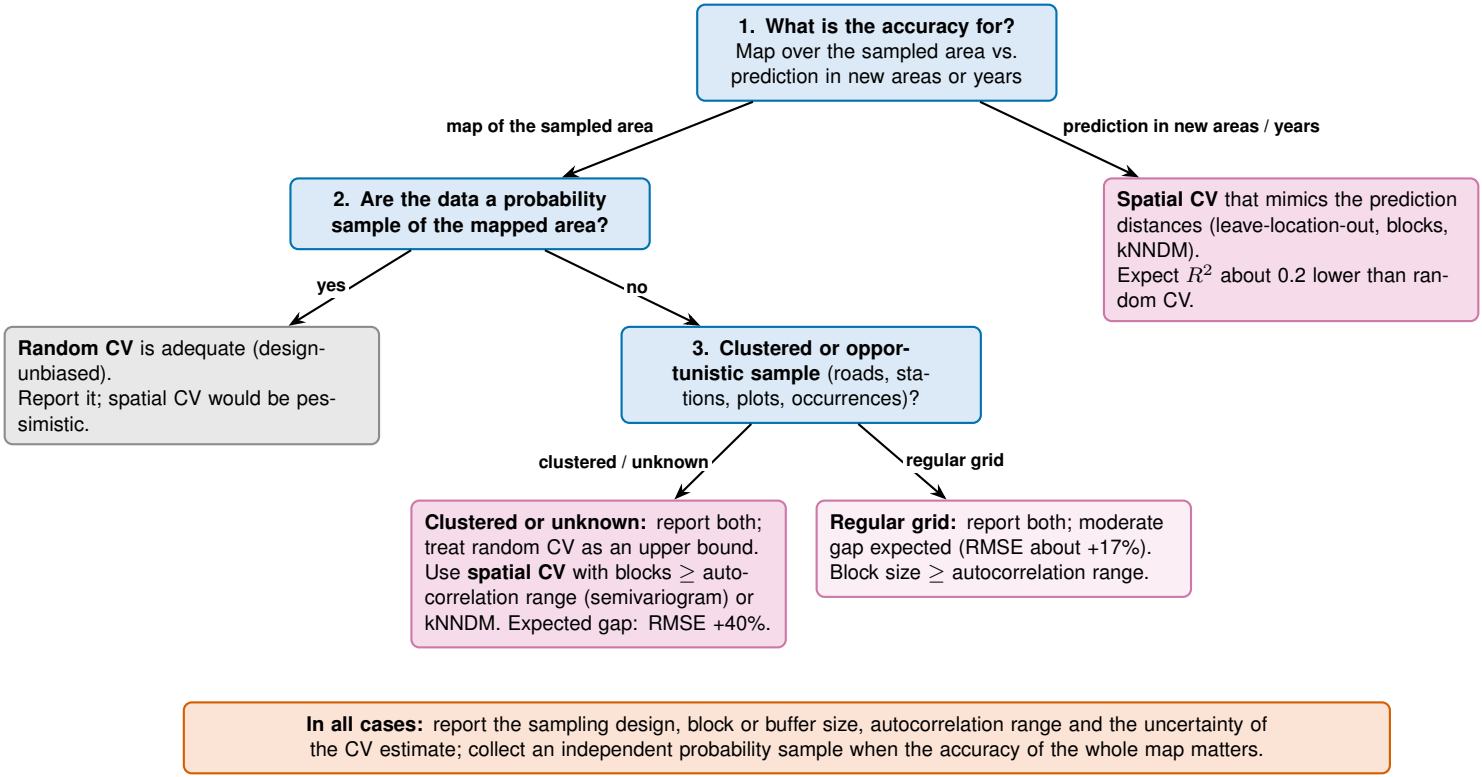
1. What is the accuracy for?
Map over the sampled area vs. prediction in new areas or years
map of the sampled area
prediction in new areas / years
2. Are the data a probability sample of the mapped area?
Spatial CV that mimics the prediction distances (leave-location-out, blocks, kNNDM).
Expect R^2 about 0.2 lower than random CV.
yes
no
Random CV is adequate (design-unbiased).
Report it; spatial CV would be pessimistic.
3. Clustered or opportunistic sample (roads, stations, plots, occurrences)?
clustered / unknown
regular grid
Clustered or unknown: report both; treat random CV as an upper bound. Use spatial CV with blocks ≥ autocorrelation range (semivariogram) or kNNDM. Expected gap: RMSE +40%.
Regular grid: report both; moderate gap expected (RMSE about +17%). Block size ≥ autocorrelation range.
In all cases: report the sampling design, block or buffer size, autocorrelation range and the uncertainty of the CV estimate; collect an independent probability sample when the accuracy of the whole map matters.